Chapter 2: Some non-parametric procedures — Exercises

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 2 covers non-parametric (distribution-free) methods for summarising survival data: estimating the survivor, hazard and cumulative hazard functions, estimating the median and other percentiles, and comparing two or more groups with the log-rank, Wilcoxon and Peto–Peto tests. These exercises emphasise interpretation and application rather than the underlying derivations.

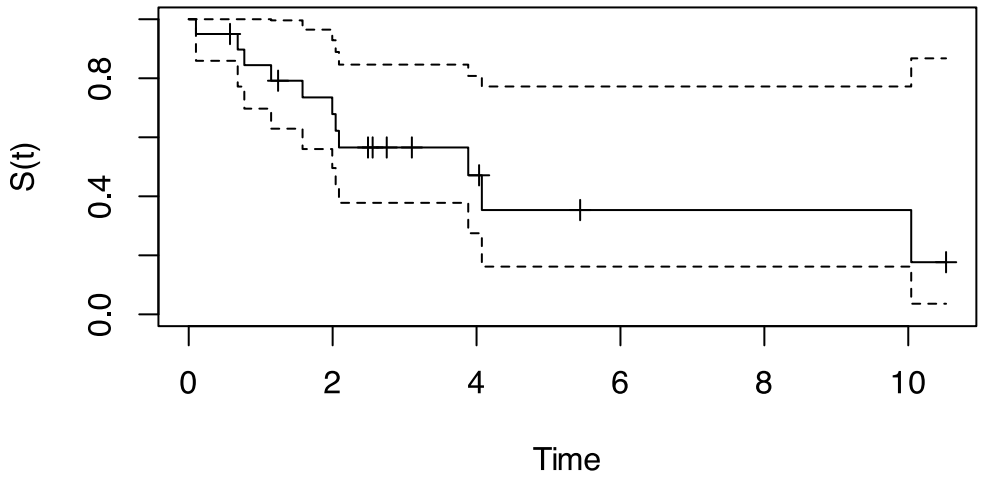
1 Estimating the survivor function (Book §2.1–2.2)

1.1 The Kaplan–Meier curve

Type: Interpret figure

Builds on: Chapter 1

The figure shows a Kaplan–Meier estimate of the survivor function for a single sample with some censored observations. The estimate is a step function, and a number of time points are marked with small tick marks.



(a) Why does the estimate drop only at observed event times, and not at the censored times?
(b) What information do the tick marks convey?
(c) The largest observation is censored. What does this imply about the estimate beyond that time?

1.2 Computing a Kaplan–Meier estimate

Type: Concept & notation

Six patients are followed; their times (months) to an event, with censored times marked with "+", are:

$$3{,}4+{,}5{,}5{,}7{,}10+$$

(a) List the distinct event times and the number at risk just before each one.
(b) Compute the Kaplan–Meier estimate of $S(t)$ at each event time.
(c) What is the value of the estimate beyond 10 months?

1.3 Life table, Kaplan–Meier and Nelson–Aalen

Type: Method selection

(a) When is the life-table (actuarial) estimate of the survivor function the appropriate choice, and when would you prefer the Kaplan–Meier estimate?
(b) How do the Kaplan–Meier and Nelson–Aalen estimates relate to each other — which is the larger, and when are they very similar?
(c) Why does the life-table estimate lose information when individual event times are actually known?

1.4 The precision of the estimate

Type: Concept & notation

From a Kaplan–Meier analysis, $\hat{S}(12 \text{ months}) = 0.65$ with a 95% confidence interval $(0.52, 0.78)$.

(a) Interpret this confidence interval in words.
(b) Explain why the standard error of $\hat{S}(t)$ — and hence the width of the confidence interval — generally increases as t increases.

2 Estimating the hazard and the median (Book §2.3–2.5)

2.1 Estimating the cumulative hazard

Type: Concept & notation

Builds on: Chapter 1

From Chapter 1, the cumulative hazard is $H(t) = -\log S(t)$.

(a) Given a Kaplan–Meier estimate $\hat{S}(t)$, how would you estimate the cumulative hazard $\hat{H}(t)$?
(b) The Nelson–Aalen estimate of $H(t)$ is the sum $\sum_j d_j/n_j$ over the event times up to t. Explain in words what this sum accumulates.
(c) If a plot of the estimated cumulative hazard is roughly a straight line, what does this suggest about the hazard function?

2.2 Estimating the median survival time

Type: Concept & notation

A Kaplan–Meier estimate gives the following values of the survivor function:

t	10	20	30	40
$\hat{S}(t)$	0.72	0.55	0.41	0.30

(a) What is the estimated median survival time, and how is it read off the estimate?
(b) In a different study, the estimate never falls below 0.5 during the whole period of follow-up. What does this mean for estimating the median, and what could you report instead?

2.3 Computing a Nelson–Aalen estimate

Type: Concept & notation

Builds on: Chapter 1

Seven patients are followed until an event (in months), with a censored time marked "+":

$$3, 5, 5, 8+, 10, 12, 15$$

(a) List the distinct event times and the number at risk just before each one.
(b) Compute the Nelson–Aalen estimate of the cumulative hazard, $\hat{H}(t)$, at each event time.

(c) Compute the Kaplan–Meier estimate $\hat{S}(t)$ at the same event times, and compare it with $\exp\left\{-\hat{H}(t)\right\}$. Which is the larger?
(d) What is the estimated median survival time?

3 Comparing two or more groups (Book §2.6–2.9)

3.1 Choosing a test for two groups

Type: Method selection

You want to compare the survival experience of two treatment groups. Which of the log-rank, Wilcoxon and Peto–Peto tests would you use in each situation, and why?

(a) The two estimated survivor functions do not cross, and proportional hazards seems reasonable.
(b) You are mainly interested in differences in survival early on, when most events occur.
(c) The pattern of censoring differs between the two groups.

3.2 The log-rank test and proportional hazards

Type: Assumptions

Builds on: Chapter 1

(a) In your own words, what does *proportional hazards* mean?
(b) Why is "the two estimated survivor functions do not cross" only a necessary, not a sufficient, check for proportional hazards?
(c) Using $S(t) = \exp\{-H(t)\}$ from Chapter 1, explain why proportional hazards implies that the two survivor functions never cross.

3.3 Interpreting a log-rank test

Type: Interpret R output

The output below compares the survival times of women with breast cancer, grouped according to whether a section of the tumour was positively or negatively stained.

```
survdiff(Surv(time, status) ~ stain)

                  N Observed Expected (O-E)^2/E (O-E)^2/V
stain=positive   13        5     9.57      2.18      3.52
stain=negative   32       20    15.43      1.35      3.52

 Chisq= 3.5  on 1 degrees of freedom, p= 0.06
```

(a) State the null hypothesis being tested.
(b) For the positively stained group, interpret the observed (5) and expected (9.57) numbers of deaths.

(c) What are the value of the test statistic, its degrees of freedom and the p-value? What do you conclude?

3.4 Computing the log-rank statistic

Type: Risk set reasoning

Two groups of patients are followed until an event (in months); there is no censoring:

Group A: 5,9,13,17 Group B: 2,4,7,11

(a) List the ordered event times, and for each give the numbers at risk in groups A and B.
(b) For group A, compute the observed number of deaths O_A and the expected number $E_A = n_A d/(n_A + n_B)$ at each event time, and hence the totals O_A and E_A.
(c) The log-rank statistic is $(O_A - E_A)^2/V$, where

$$V = \sum_j \frac{n_{Aj} n_{Bj} d_j (n_j - d_j)}{n_j^2 (n_j - 1)}.$$

```
Compute the statistic and compare it with the $\chi^2$ distribution on one degree
of freedom.
```

(d) Interpret the result.

3.5 Stratifying a comparison

Type: From research question to model

A multicentre trial compares two chemotherapy treatments for lung cancer. Survival times are available from several centres, and age is also recorded.

(a) Why might you stratify the comparison by centre rather than simply ignore centre?
(b) What does a stratified log-rank test provide that separate (per-centre) tests do not?
(c) In words, what is the null hypothesis of the stratified test?

3.6 A misleading comparison

Type: What's wrong here?

A researcher compares two treatment groups by computing, for each group, the proportion of patients still alive at the end of the study, and then compares these two proportions with a chi-squared test. Patients entered the study at different times and were followed for different lengths of time.

Explain what is wrong with this approach.

4 Mini-lab: Kaplan–Meier and the log-rank test in R

4.1 Comparing two groups in R

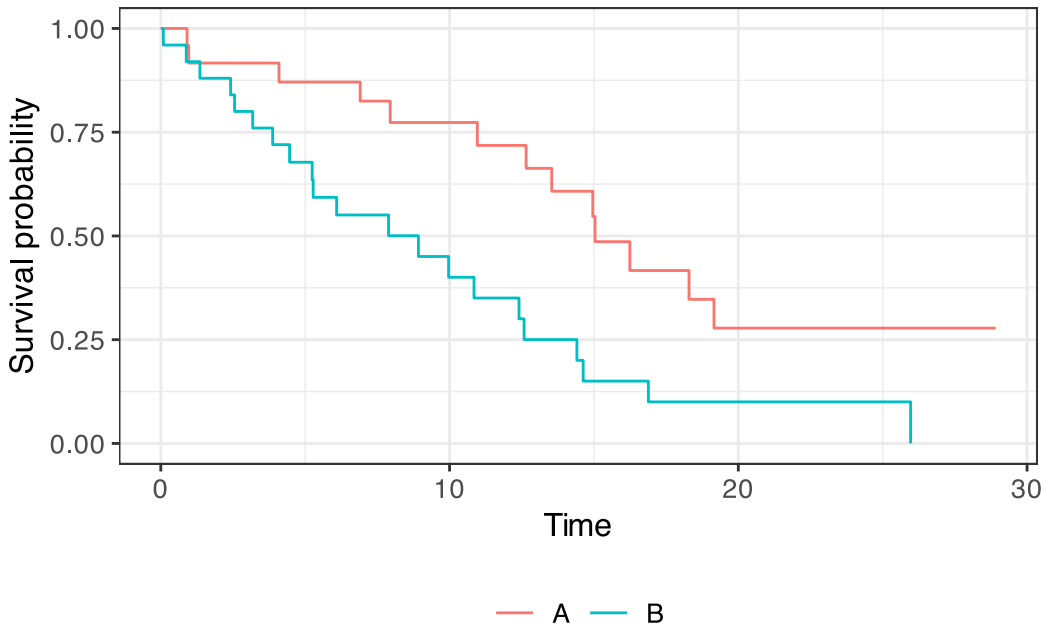
Type: Mini-lab

Run the code below. It simulates two groups of 25 patients (Group A with a lower hazard than Group B) with independent censoring, fits Kaplan–Meier curves, and runs a log-rank test.

```
set.seed(7)
n <- 25
time  <- c(rexp(n, 0.05), rexp(n, 0.10))   # Group A then Group B
cens  <- runif(2 * n, 0, 40)
obs   <- pmin(time, cens)
event <- as.numeric(time <= cens)
grp   <- factor(rep(c("A", "B"), each = n))
d <- data.frame(time = obs, status = event, group = grp)

library(survival)
library(ggsurvfit)

survfit2(Surv(time, status) ~ group, data = d) |>
  ggsurvfit() +
  labs(x = "Time", y = "Survival probability")
```




```
survdiff(Surv(time, status) ~ group, data = d)
```

```
Call:
survdiff(formula = Surv(time, status) ~ group, data = d)
```

```
         N Observed Expected (O-E)^2/E (O-E)^2/V
group=A 25       13     20.6      2.83      7.49
group=B 25       21     13.4      4.36      7.49

 Chisq= 7.5  on 1 degrees of freedom, p= 0.006
```

(a) From the plot, which group appears to have better survival? Estimate each group's median survival time.

(b) What is the value of the log-rank test statistic, its degrees of freedom and its p-value? State the null hypothesis and your conclusion.

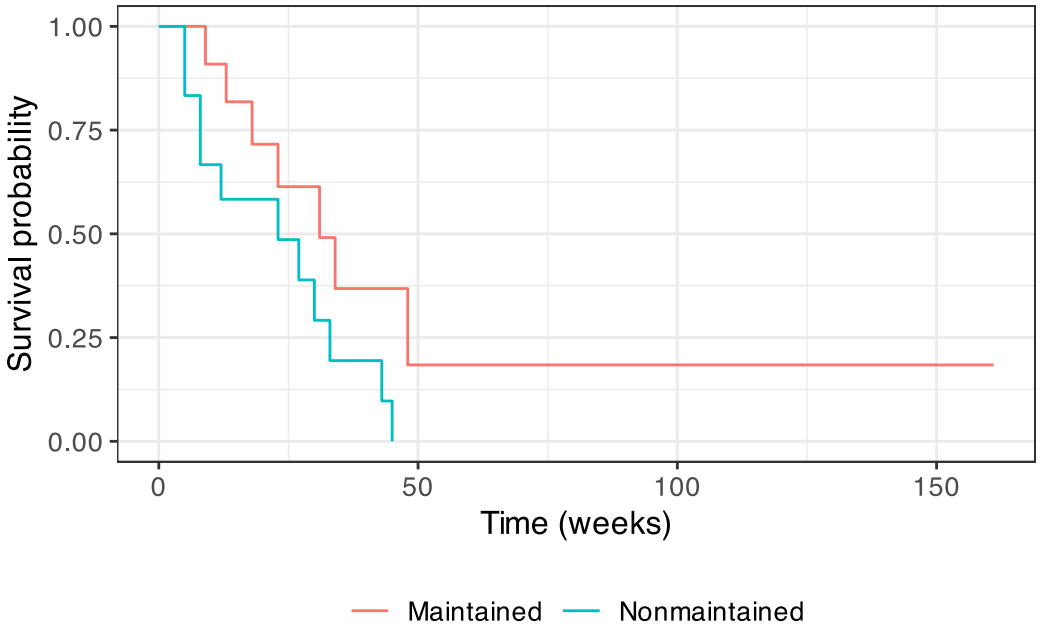
4.2 Analysing a real dataset: acute myeloid leukaemia

Type: Mini-lab

The `aml` dataset (in the `survival` package) contains survival times for 23 patients with acute myeloid leukaemia, grouped by whether they received maintenance chemotherapy (`x`: "Maintained" or "Nonmaintained"). Run the code below to plot the Kaplan–Meier curves and run a log-rank test.

```r
library(survival)
library(ggsurvfit)
data(aml)

survfit2(Surv(time, status) ~ x, data = aml) |>
  ggsurvfit() +
  labs(x = "Time (weeks)", y = "Survival probability")
```




```
survdiff(Surv(time, status) ~ x, data = aml)
```

```
Call:
survdiff(formula = Surv(time, status) ~ x, data = aml)

                N Observed Expected (O-E)^2/E (O-E)^2/V
x=Maintained    11        7    10.69      1.27       3.4
x=Nonmaintained 12       11     7.31      1.86       3.4

 Chisq= 3.4  on 1 degrees of freedom, p= 0.07
```

(a) How many patients are in each group, and how many events (deaths) occurred in each?
(b) From the plot, which group appears to have better survival? Report each group's estimated median survival time.
(c) Report the log-rank statistic, its degrees of freedom and p-value. State the null hypothesis and your conclusion.